Yusuf Roohani

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Education

2020–24	**Ph.D.**, Biomedical Data Science, Stanford University	GPA: 4.10/4.25
	Dissertation Title: *Engineering Cells using Artificial Intelligence*	
	Advisors: Jure Leskovec, Stephen Quake	
	Committee: Stanley Qi, Barbara Engelhardt, Daphne Koller, Silvana Konermann	
2014-15	**M.S.**, Mechanical Engineering, Carnegie Mellon University	GPA: 4.00/4.00
2009-13	**B.Tech.** Mechanical Engineering, Vellore Institute of Technology	GPA: 8.81/10.00

Work Experience

Arc Institute, Palo Alto, CA

Jan-Jun 2026 **Associate Director, Machine Learning**

2024-2025 **Group Lead, Machine Learning**

Led development of AI model of cell state to predict the outcomes of experiments
Closed the loop with experimental teams to enable AI-guided data generation
Built a new interdisciplinary team of 10+ ML research scientists, engineers
3 papers published at Cell, 1 under review at Nature.
Member of Arc Institute AI Advisory Board

Stanford University, Stanford, CA

2024-2025 **Visiting Scholar: Department of Computer Science**

2020-2024 **PhD Student: Department of Biomedical Data Science**

Foundation models for cell biology + LLM-driven agents for experiment design.

GlaxoSmithKline, Cambridge, MA

2019-2020 **Manager, Machine Learning Engineer**

2017-2019 **Investigator**

2016-2017 **Data Scientist**

Led a cross-disciplinary team to biologically profile a 2M+ compound collection using multi-modal datasets and high-throughput screening.

Merrimack Pharmaceuticals, Cambridge, MA

2015-2016 **Computational Modeler Intern**

Developed dynamic system models to simulate signaling networks in cancer cells.

Publications

Google Scholar Page † → Equal contribution

Selected Journal Articles

J1. Adduri, A., Gautam, D., Bevilacqua, B., ... & **Roohani, Y.** Predicting cellular responses to perturbation across diverse contexts with STATE. *Cell* (2026).

J2. Huang, K., Zhang, S., Wang, H., Qu, Y., Lu, Y., **Roohani, Y.**, Li, R., Qiu, L., Li, G., Zhang, J., *et al.* Autonomous biomedical research with an artificial intelligence agent. *Science* (2026).

J3. Rosen Y.[†], **Roohani Y.**[†], Agarwal, A., Samotorčan, L., ... Quake, S. R. & Leskovec, J. Universal cell embeddings: A foundation model for cell biology. *Nature* (2026).

J4. Youngblut, N., Carpenter, C., ..., Goodarzi, H.[†] & **Roohani, Y.** [†]. scBaseCount: an AI agent-curated, uniformly processed single cell data repository. *Cell* (2026).

J5. **Roohani, Y.**, Hua, T. J., Tung, P.-Y. & ... Burke, D. Virtual Cell Challenge: Toward a Turing test for the virtual cell. *Cell* (2025).

J6. Bunne C.[†], **Roohani Y.**[†], Rosen Y.[†], Gupta, A., *et al.* How to build the virtual cell with artificial intelligence: Priorities and opportunities. *Cell* (2024).

J7. **Roohani, Y.**, Huang, K. & Leskovec, J. Predicting transcriptional outcomes of novel multigene perturbations with GEARS. *Nature Biotechnology* (2024).

J8. Rosen, Y.[†], Brbić, M.[†], **Roohani Y.**[†], Swanson, K., Li, Z. & Leskovec, J. Toward universal cell embeddings: integrating single-cell RNA-seq with SATURN. *Nature Methods* (2024).

J9. **Roohani, Y.**, ..., Robinson, A. L. & Adams, P. J. Impact of natural gas development in the Marcellus and Utica shales on regional ozone and $PM_{2.5}$. *Atmospheric Environment* (2017).

Peer-reviewed Conference Proceedings

C1. **Roohani Y.**[†], Lee A.[†], Huang Q.[†], ..., Marson, A., Liang, P. & Leskovec, J. BioDiscoveryAgent: An AI Agent for Designing Genetic Perturbation Experiments. *ICLR* (2025).

C2. Nilforoshan, H.[†], Moor, M.[†], **Roohani, Y.**, Chen, Y., Šurina, A., Yasunaga, M., Oblak, S. & Leskovec, J. Zero-shot causal learning. *NeurIPS* (2023).

Preprints

W1. Dong, M., Adduri, A., Gautam, D., Carpenter, C., Shah, R., Ricci-Tam, C., ... & **Roohani, Y.** STACK: In-Context Learning of Single-Cell Biology. *Under review at Science* (2025).

W2. Magnusson J.[†], **Roohani Y.**[†], ... & Qi, L. S. PreciCE: Precision engineering of cell fates via multi-gene control of transcriptional networks. *Under review at Mol. Cell* (2024).

W3. Bommasani, R., Hudson, D. A., Adeli, E., Altman, R., Arora, S., **Roohani, Y.**, ., ... & Liang, P. On the opportunities and risks of foundation models. *arXiv* (2021).

Patents

P1. Adduri, A., **Roohani, Y.**, Goodarzi, H. & Burke, D. A Scalable Approach for Learning Perturbation Effects Over Sets of Cells 63/823,201. *U.S. Patent Application* (Filed June 2025).

Selected Invited Talks

2026	Generative and Agentic AI for Biology, ICML, Seoul, South Korea
2026	Google DeepMind, London, UK
2026	Gordon Research Conference, Single Cell Genomics, Les Diablerets, Switzerland
2026	Fred Hutchinson Cancer Center, Seattle, WA
2026	Broad Institute (Models, Inference and Algorithms Seminar), Cambridge, MA
2025	Novartis Advances in Foundation Models Workshop, Cambridge, MA
2025	Columbia University Biomedical Informatics Seminar, New York, NY
2025	UCLA Bioengineering Seminar, Los Angeles, CA
2023, 2025	Genentech Research and Early Development, South San Francisco, CA
2025	Stanford AI & Biomedicine Seminar, Stanford, CA
2025	Xaira Therapeutics, South San Francisco, CA
2025	Recursion Pharmaceuticals, Salt Lake City, UT
2023, 2025	Society for Lab Automation and Screening Conference, San Diego, CA
2024	US Science Day, UCB Pharma, Cambridge, UK
2023, 2024	Stanford Graph Learning Workshop 2024, Stanford, CA

Advisory & Board Roles

2026	**Pfizer** Foundation Model Scientific Advisory Board
2026	**Arc Institute** AI Advisory Board
2026	**GC Therapeutics** Scientific Advisor

Awards & Honors

2024	**Best Poster** ICLR 2024 Machine Learning for Genomics Exploration Workshop
2023	**Best Poster** Top 3 out of 200 at Stanford BioX Interdisciplinary Initiatives ($500)
2023	**Innovation Award** Society for Lab Automation and Screening ($10K)
2022	**Best Poster** Intelligent Systems for Mol. Biology (ISMB) (Machine Learning Track)
2022	**Best Poster** Single Cell Genomics meets Data Science, Munich ($500)
2019	**Full PhD funding** Awarded by GSK, including tuition + full-time pay ($1M+)
2018	**GSK Exceptional Science Award** Deep learning for cellular images ($17K)

Student Mentorship

2023-2026	Mentored 10+ BS and 5 PhD students from Stanford, UC Berkeley, Penn, and Yale.

Reviewing

Nature, Nature Biotechnology, Nature Aging, Nature Methods, NeurIPS, PLOS Biology, PLOS Computational Biology, Nucleic Acids Research, MICCAI

Software

GEARS (400+ stars), UCE (300+ stars), BioDiscoveryAgent (100+ stars), STATE (700+ stars)